

Oklahoma's Hidden Meteor Crater Is Younger Than Scientists Thought

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Deep under the town of Ames, Oklahoma, there is a giant **meteor crater**. It stretches for miles underground. Now scientists say it is almost 100 million years younger than they believed.

For years, people thought the **crater** formed about 467.5 million years ago. That would have made it part of a group of big **meteor** hits across North America. Some scientists even wondered if Earth once had a ring of space rocks, like Saturn.

A team from The University of Texas at Austin studied tiny crystals called **zircons** in rock changed by the crash. The crystals showed the **meteor** hit about 370 million years ago.

Why was the old date wrong? It came from **fossil** teeth of an eel-like animal. Lead scientist Elizabeth Catlos said the teeth were probably already old when the **meteor** hit, and the crash mixed them into the rocks.

Word Treasure

meteor -- A space rock that falls toward Earth.

crater -- A big hole made when something crashes into the ground.

zircon -- A tiny crystal found in rocks that can help tell their age.

fossil -- Remains of a living thing from long ago, saved in rock.

Break it down

A giant crater hides under Oklahoma
The old idea about its age
How tiny crystals gave a new date
Why the old date was wrong

Why it matters

Getting the dates right helps scientists understand what happened on Earth long ago.

Test what you remember

1. Where is the meteor crater?

- ☐ (A) On the Moon
- ☐ (B) In the ocean near Hawaii
- ☐ (C) Under Paris, France
- ☐ (D) Under Ames, Oklahoma

2. How much younger is the crater than scientists thought?

- ☐ (A) Exactly one million years
- ☐ (B) Almost 100 million years
- ☐ (C) It is older, not younger
- ☐ (D) About 10 years

3. What did some scientists wonder Earth once had?

- ☐ (A) A ring of space rocks, like Saturn
- ☐ (B) Two suns
- ☐ (C) No oceans
- ☐ (D) A second moon made of ice

4. What tiny things did the team study?

- ☐ (A) Crystals called zircons
- ☐ (B) Grains of sugar
- ☐ (C) Butterfly wings
- ☐ (D) Raindrops

5. When do scientists now think the meteor hit?

- ☐ (A) About 370 million years ago
- ☐ (B) Last year
- ☐ (C) About 5,000 years ago
- ☐ (D) About 467.5 million years ago

6. Where did the old date come from?

- ☐ (A) Fossil teeth of an eel-like animal
- ☐ (B) A history book
- ☐ (C) A tree's rings
- ☐ (D) A photo from space

Answer key (try the quiz first!): 1: D 2: B 3: A 4: A 5: A 6: A

MY THOUGHTS (at least 20 words):